Aim: Study of Bluetooth communication using mobile devices(pairing , range, testing)

Bluetooth & Piconets

- Definition: A Piconet is the most basic network cluster formed by Bluetooth devices. It is an ad-hoc network consisting of one primary "Master" device and up to seven active "Slave" devices communicating on the same channel.
- Features:
    - o Central Control: The Master device acts as the network clock, dictating the exact frequency-hopping schedule for all connected slaves.
    - o Dynamic Roles: Devices can switch roles dynamically; for example, a phone can be a slave to a laptop but a master to a smart watch.
    - o Parked Devices: Beyond the 7 active slaves, up to 255 additional devices can stay connected in a low-power "parked" state, waiting for the master to wake them up.
- Advantages:
    - o Perfect Synchronization: Because one clock rules the network, data packets never collide within the cluster.
    - o Instant Set-up: Devices form these networks automatically on the fly without needing manual network configuration.

Scatternets

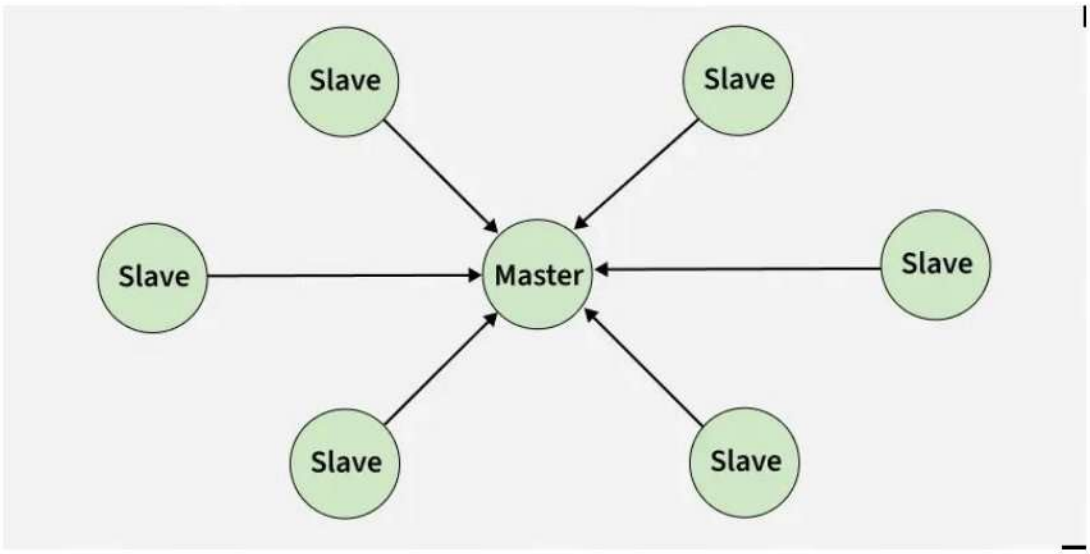
- Definition: A Scatternet is a larger network formed by chaining two or more Piconets together.
- Features:
    - o Bridge Devices: Piconets link up via a "bridge device"—a node that acts as a Slave in one piconet and a Master (or Slave) in another.
    - o Time-Sharing: The bridge device splits its time, jumping back and forth between the frequency schedules of both piconets to move data across.
    - o Expanded Scalability: It breaks the 7-device limit of a single piconet, allowing data to hop across a much longer chain of hardware.
- Advantages:
    - o Extended Range: Data can travel across a whole room or floor by hopping through bridge devices, bypassing Bluetooth’s strict 10-metre limit.

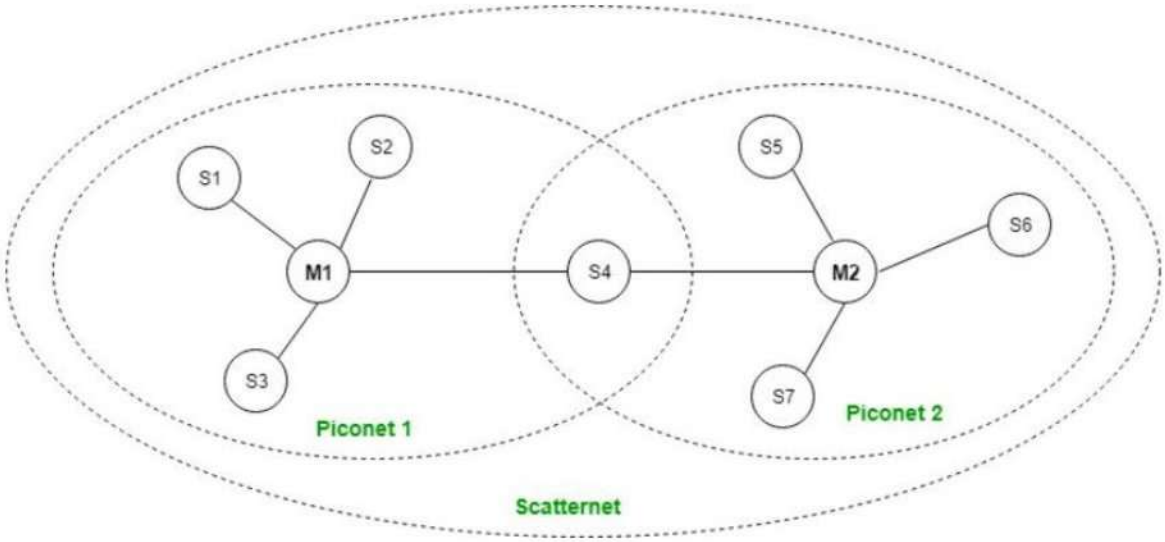
- Seamless IoT Web: It creates an inexpensive, self-healing mesh of devices without needing an internet router.

The Pairing & Bonding Process

- Definition: Pairing is the initial security handshake where two strangers exchange keys to build trust. Bonding is storing those keys in long-term memory so they can auto-connect in the future.
- Features:
  - Inquiry & Paging: The phone first broadcasts an "inquiry" to find nearby open devices, then sends a targeted "page" to knock on the chosen device's door.
  - Secure Simple Pairing (SSP): Devices automatically generate a random 6-digit pin code on both screens for the user to confirm, blocking hackers from intercepting the link.
- Advantages:
  - Bulletproof Privacy: Encryption ensures your wireless headphones cannot be hijacked or listened to by someone sitting next to you.
  - One-Time Setup: Once bonded, you never have to repeat the pairing hassle; your phone connects instantly the moment you turn on your earbuds.

Reading No.	Distance (Metres)	Measured RSSI (dBm)	Visual/Audio Performance Observations
1	**0.5 m**	-42 dBm	File transfer is near-instantaneous. Signal is perfectly stable.
2	**2.0 m**	-55 dBm	Fast data transfer. No noticeable latency.
3	**5.0 m**	-68 dBm	Stable connection. Audio streams smoothly without lag.
4	**10.0 m**	-81 dBm	File transfer slows down noticeably. Reaching the fringe of Class 2 range.
5	**12.0 m**	-89 dBm	Connection is highly unstable. File transfer fails or stalls completely.
6	**> 14.0 m**	*Disconnected*	Signal drops entirely. Phone searches for the device.


Slave
Slave
Slave
Master
Slave
Slave
Slave


S1
S2
M1
S3
S4
S5
M2
S6
S7
Piconet 1
Piconet 2
Scatternet